EL QUESTIONS

LQQNPG

Part – I

Choose the Correct answer (1 Mark)

1. Density of pure water is _____ .
   a. 2000 kg/m^3 b. 1000 kg/m^3
   c. 1500 kg/m^3 d. 2500 kg/m^3
2. The unit for surface tension is _____.
   a. N/mm b. N/mm^2
   c. N/m d. N/cm
3. The pressure happens when a fluid is at rest is _____.
   a. Static Pressure
   b. Gauge Pressure
   c. Absolute Pressure
   d. Vacuum Pressure
4. In steady flow, the quantity of flowing liquid is _____.
   a. Non Constant b. unsteady
   c. constant d. varying
5. An Orifice is used to measure the _____ of liquid.
   a. Velocity b. Head loss
   c. Pressure d. Discharge
6. Head loss due to friction in a pipe is denoted by _____.
   a. h_f b. f_h
   c. Q d. v

Part – II

Answer in one or two sentences (3 Marks)

7. List any four properties of fluids.
8. Write short note on Density.
9. Define Pascal's Law?
10. List the types of Pressure.
11. List out the types of liquid flow.
12. Define – Orifice.
13. Write the types of mouthpieces according to its shape.
14. Write the Darcy's formula to find out loss of head.
15. What do you mean by pump?

Part – III

Answer in brief (5 Marks)

16. Write briefly about
    (a) Fluid pressure.
    (b) Intensity of pressure.
17. State the Bernoulli's Theorem or Energy equation.
18. Write briefly about Vena contracts with simple sketch.
19. Write briefly about Hydraulic gradient line with sketch.

Part – IV

Answer in detail (10 Marks)

20. Water flows through a pipe of 250 mm diameter and 200 m long with a velocity of flow 2.75 m/s. Find the loss of head due to friction. Take friction factor (f) as 0.003
21. Determine the loss of head in a pipe line 200 mm diameter and 450 m long. The discharge of water through the pipe is 0.255 m^3/s. Take the value of friction factor as 0.002.
22. Water flows through a pipe line 150 mm diameter and 400 m long. The velocity of flow of water in the pipe is 2.5 m/s. Find out the loss of head in the pipe by taking Chezy's constant as 40.

Answers

1. (b) 2. (c) 3. (a) 4. (c) 5. (d) 6. (a)

U7.indd 153 12/11/2019 5:34:07 PM